

NexSuite Product Overview

Sysnex Labs

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NexSuite Product Overview

The Modern SysML v2 Language Server for Enterprise MBSE

Executive Summary

NexSuite is a production-ready, enterprise-grade SysML v2 language server that brings modern software development practices to Model-Based Systems Engineering (MBSE). Built with Rust for exceptional performance and integrated natively into Visual Studio Code, NexSuite delivers 10x faster performance than legacy MBSE tools while providing AI-powered assistance, Git-native workflows, and comprehensive industry compliance automation.

Key Benefits

- **10x Performance Advantage:** Sub-50ms LSP response times vs. 200-500ms in traditional tools
 - **Zero Learning Curve:** Native VS Code integration leverages 100M+ existing users
 - **AI-First Design:** Built-in GitHub Copilot and Claude Code support (40-60% productivity boost)
 - **100% SysML v2 Compliance:** Future-proof standard support with full KerML implementation
 - **Industry Compliance Automation:** ASPICE, ISO 26262, ISO 15288 work product generation
 - **Cost Effective:** Free tier available vs. \$5K-\$50K+ enterprise licensing
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Product Architecture

NexSuite follows a Rust-Analyzer-inspired layered architecture with query-based incremental computation:

VS Code / Cursor Extension	
LSP Server (Tower-LSP)	
IDE Features - Completion - Navigation - Diagnostics - Code Actions	Domain Extensions - UVL Variability - VSS Integration - YAML Architecture - Trade Studies
High-Level IR (HIR) & Type System	
Salsa Incremental Database	
Lossless Syntax Layer (Never-Fail Parser)	

Technical Foundation

- **70 Specialized Crates:** 271K+ lines of production Rust code
- **3,299+ Tests:** 99.9% pass rate ensuring reliability
- **Salsa Database:** Automatic incremental computation and memoization
- **Never-Fail Parser:** Robust error recovery for incomplete code
- **Memory Safe:** Rust eliminates 70% of common vulnerabilities

Core LSP Features (14/14 Complete)

All standard Language Server Protocol features fully implemented and production-ready:

Feature		Performance	Description
Code Completion		<30ms	Context-aware suggestions with 14 completion types
Go-to-Definition		<5ms	Cross-file navigation with URI resolution
Find References		<50ms	Multi-file reference search
Hover Information		<50ms	Rich markdown with SysML v2 semantics
Diagnostics		<5ms (cached)	16 collectors for real-time error reporting
Semantic Highlighting		~80ms	43 token types (19 LSP + 24 SysML v2)
Code Actions		<50ms	8 providers with 14 action kinds
Rename Symbol		<100ms	Multi-file with conflict detection
Formatting		~50ms	Document and range formatting
Folding Ranges		<50ms	Nested structure support
Document Symbols		<50ms	Hierarchical symbol outline
Workspace Symbols		<150ms	Global search with 5 filter types
Inlay Hints		<50ms	Type and parameter hints
Signature Help		<50ms	Function signatures with parameters

Advanced LSP Features

- **Call Hierarchy:** Incoming/outgoing calls visualization
 - **Type Hierarchy:** Supertype/subtype chains
 - **Document Links:** File paths, imports, external references
 - **Code Lens:** Reference counts, implementation counts
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Domain-Specific Extensions

UVL Variability Management

Complete feature variability modeling with bidirectional sync:

- **SAT/SMT Solving:** Configuration validation with varisat and Z3 integration
- **Dead Feature Detection:** Identify features that can never be selected
- **Model Derivation:** 150% to 100% model transformation
- **Trade Study Integration:** Generate trade studies from valid configurations
- **Live Preview:** Real-time visual feedback on feature states

15 LSP Commands | 5 Crates | 2,630 Lines | 21 Tests

VSS Integration (Vehicle Signal Specification)

- Import and sync vehicle signal specifications
- Bidirectional transformation between VSS and SysML v2
- 11 LSP commands for VSS operations

YAML Architecture Parser

- ADL/SUDL (Architecture Description Language) support
- Transformation to SysML v2 models
- Loader and code generation capabilities

Python Bindings (85% Complete)

- FFI for Python automation
 - Model manipulation API
 - 15 modules for CRUD operations
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Industry Compliance Automation

ISO/IEC 15288 (83% Complete)

Foundation for all compliance variants with 6 frameworks:

- System Analysis Framework (95%)
- Risk Management Framework (85%)
- Decision Management Framework (95%)
- Design Definition Framework (80%)
- Implementation Framework (75%)
- Project Planning Framework (80%)

ASPICE (40% Complete - 9/22 Features Production-Ready)

16,295+ lines | 100+ tests | 15 of 16 processes supported

Production-ready features: - Requirements engineering and traceability - ASIL classification and tracking - Change impact analysis - Configuration management - Constraint validation (288 rules, 95-100% precision) - Documentation generation - Model analytics and metrics

ISO 26262 Integration

- ASIL tracking and classification
- ASIL decomposition validator (Part 9)
- HARA (Hazard Analysis and Risk Assessment) integration
- Safety requirement traceability

Platform Support

Operating Systems

Platform	Architecture	Status
macOS	x86_64, ARM64 (Apple Silicon)	 Production
Linux	x86_64, ARM64	 Production
Windows	x86_64	 Production
Raspberry Pi 5	ARM64	 SaaS Server

IDE Integrations

IDE	Status	Method
Visual Studio Code	 Production	Native extension
Cursor	 Production	VS Code compatible
IntelliJ IDEA	Community contribution	LSP protocol
Vim/Neovim	Community contribution	LSP protocol
Web Browser	 Working	WASM + WebSocket LSP

Deployment Options

- 1. VS Code Extension** - VSIX package distribution - Marketplace-ready (v0.33.0) - 35+ UI components - Full LSP integration
- 2. Tauri Desktop App** (In Development) - Standalone application - No VS Code required - Cross-platform (Windows, macOS, Linux) - Full offline capability
- 3. Web IDE (SaaS)** - Browser-based IDE - REST API backend (18 endpoints) - WebSocket LSP connection - Real-time collaboration ready

Performance Benchmarks

All targets exceeded by significant margins:

Operation	Target	Achieved	Improvement
Code Completion	<50ms	<30ms	40% better
Go-to-Definition	<50ms	<5ms	90% better
Find References	<100ms	<50ms	50% better
Diagnostics (cached)	<150ms	<5ms	97% better
Formatting	<200ms	~50ms	75% better
Full Model Load	<2s	<250ms	87.5% better
Memory Usage	<150MB	~120MB	20% better

Technical Specifications

SysML v2 Support

- **100% SysML v2 Release Standard Compliance**
- Full KerML (Kernel Modeling Language) support
- All SysML v2 language constructs
- Standard Model Library (402 files)
- Parametric modeling
- Constraint expressions (OCL 2.5)

Technology Stack

Backend (Rust) - Rust nightly toolchain - Salsa 0.24 (incremental computation) - Tower-LSP 0.20 (LSP protocol) - Axum 0.7 (REST API server) - Tokio (async runtime) - varisat + Z3 (constraint solving)

Frontend (TypeScript/React) - TypeScript 5.x - React 18 - Monaco Editor - React Flow (diagram editor) - 35+ shared UI components

Git & CI/CD Integration

Native Git Support - Full Git workflow integration - Visual diff and merge conflict resolution - LSP validation during merge - Pull request and code review - Zero-cost version control

CI/CD Pipeline - 4 GitHub Actions workflows - Multi-platform builds (macOS, Linux, Windows, ARM64) - 9 specialized Docker images - Automated testing (3,299+ tests) - Load testing validated (100% success rate)

Documentation & Visualization

MkDocs Generator

- User-facing documentation generation
- Workspace-level documentation
- Interactive navigation

Sphinx Integration

- Technical/API documentation
- Type information inclusion
- Code examples

6 Diagram Types

- PlantUML
- Mermaid
- D2
- DOT
- SVG
- ASCII

Requirements Manager

- Traceability matrix generation
- Validation and verification
- Gap detection

Pricing & Licensing

Tier	Features	Price	Target
Essential	Core LSP features	FREE	Individual developers
Platform-Full	All features except compliance	FREE	Small teams
Automotive	ASPICE, ISO 26262	\$2.5K-\$18K/year	Enterprise automotive

Compare to Traditional MBSE Tools: - Enterprise Architect: \$229-\$799/user one-time - Cameo Systems Modeler: \$5K+/user/year - Legacy enterprise suites: \$50K-\$200K+ for SCM systems

Why Choose NexSuite?

10x Performance

Rust-based architecture delivers sub-50ms LSP response times compared to 200-500ms in Java-based legacy tools. Real-time diagnostics, instant navigation, and smooth editing experience.

Zero Learning Curve

Native VS Code integration means your team already knows the tool. No expensive training programs or workflow disruption. Start modeling in minutes, not weeks.

AI-Powered Productivity

Built-in GitHub Copilot and Claude Code integration provides 40-60% productivity boost. AI understands SysML v2 semantics and generates models, not just code.

Future-Proof Standards

100% SysML v2 compliance ensures your models are compatible with the next generation of MBSE tools and workflows. Full backwards compatibility with migration tools.

Git-Native Collaboration

Zero-cost collaboration using Git. No expensive Teamwork Cloud (\$50K-\$200K) or centralized servers. Distributed workflows with complete audit trails.

Industry Compliance

Automated work product generation for ASPICE (20/20 types), ISO 26262 ASIL tracking, and ISO 15288 frameworks. Reduce compliance overhead by 50-70%.

Getting Started

Quick Start (5 minutes)

1. Install VS Code Extension

```
code --install-extension nexsuite.sysml-v2-lsp
```

2. Open SysML v2 File

- Create new `.sysml` file
- Start modeling with full LSP support

3. Explore Features

- Code completion (Ctrl+Space)
- Go-to-definition (F12)
- Find references (Shift+F12)
- Documentation generation

Resources

- **Documentation:** <https://docs.nexsuite.dev>
 - **Examples:** <https://github.com/nexsuite/examples>
 - **Community:** <https://discord.gg/nexsuite>
 - **Support:** support@nexsuite.dev
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Technical Support

Community Support (FREE)

- [GitHub Issues](#)
- [Discord Community](#)
- [Documentation Portal](#)
- [Video Tutorials](#)

Enterprise Support (\$5K-\$25K/year)

- Dedicated support team
 - SLA guarantees (4-hour response)
 - Custom feature development
 - On-site training and consulting
 - Priority bug fixes
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About Sysnex Labs

Sysnex Labs is pioneering the next generation of Model-Based Systems Engineering tools. Our mission is to bring modern software development practices—AI assistance, Git workflows, and exceptional performance—to systems engineering teams worldwide.

Version: 0.33.0 **Release Date:** January 2026 **Status:** Production-Ready

Contact Information

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NexSuite - Modern MBSE for Modern Teams